

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)
Question Count: 3

Total Marks: 30

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Case Study

1. Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

2. Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices.

Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

3. Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients.

Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform.

Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced

through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized

Case Study 1 : Smart Retail Inventory Management System

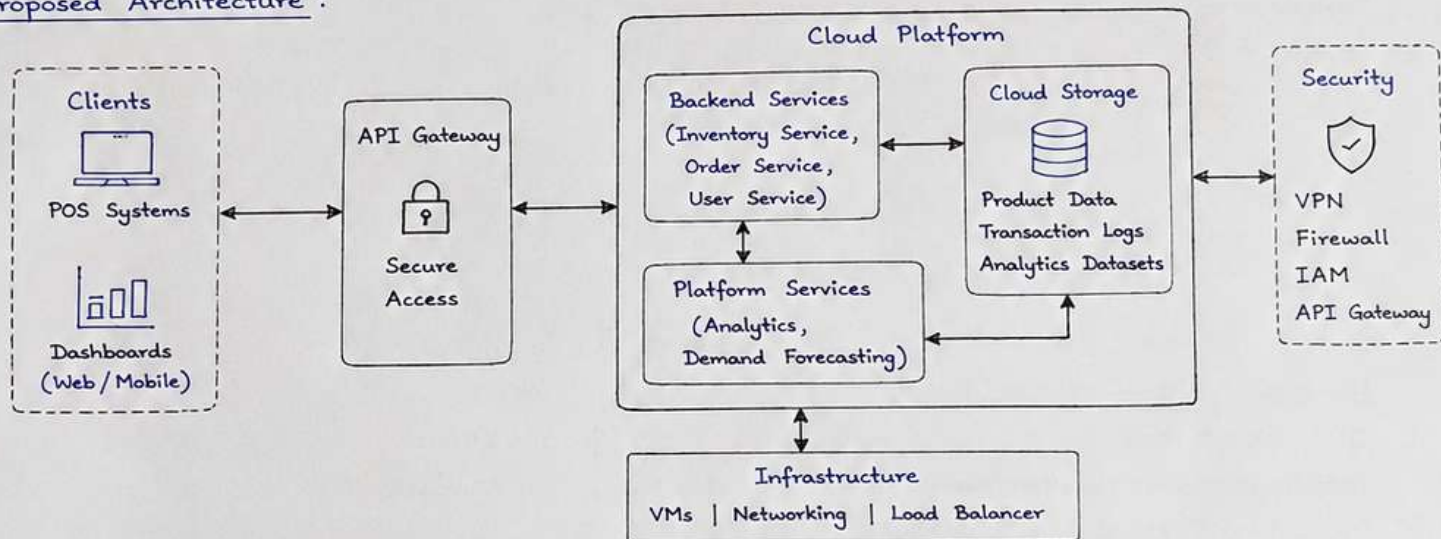
1. Introduction :

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

2. Cloud Components Used :

- Clients : POS systems, Dashboards (Web/Mobile)
- Web APIs : RESTful APIs for communication
- Cloud Storage : Product data, transaction logs, analytics datasets
- Platform Services : Analytics, Demand forecasting
- Infrastructure : Virtual Machines, Networking, Load Balancer, Security Services

3. Proposed Architecture :



4. Working / Data Flow :

Store devices (POS systems) send data via APIs to the API Gateway. The gateway authenticates and forwards requests to backend services. Backend services process the requests and store/retrieve data from Cloud Storage. Platform services analyze the data to generate insights and demand forecasts. Dashboards fetch data through APIs and display real-time reports to users.

5. Benefits :

- Real-time inventory tracking
- Better demand forecasting
- Scalable and reliable system
- Centralized data and analytics
- Cost-effective operations

6. Security :

- Data encryption (in transit & at rest)
- VPN, Firewall, API Gateway
- Role-based access control (RBAC)
- Identity and Access Management (IAM)
- Secure authentication for users

7. Challenges / Trade-offs :

- Internet dependency
- Cost management for cloud resources
- Data privacy compliance
- Integration with legacy systems

8. Conclusion :

The Smart Retail Inventory Management System integrates clients, APIs, cloud storage, platform services, and infrastructure to deliver a secure, scalable, and efficient solution for retail operations.

Case Study 2 : Cloud-Based Document Collaboration System

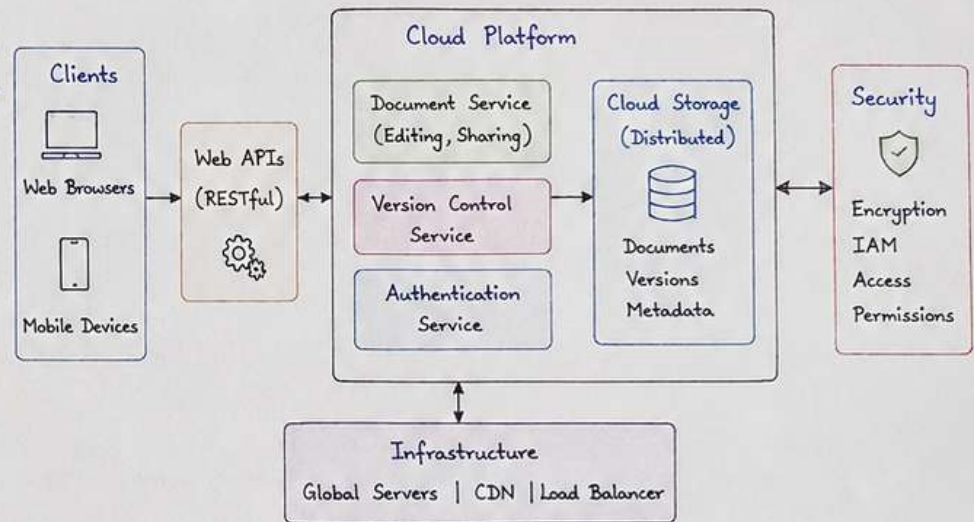
1. Introduction :

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

2. Cloud Components Used :

- Clients : Web Browsers, Mobile Devices
- Web APIs : For document synchronization
- Cloud Storage : Distributed storage with version control
- Platform Services : Authentication, Notification, Real-time Collaboration
- Infrastructure : Global Servers, CDN, Load Balancer

3. Proposed Architecture :



4. Working / Data Flow :

Users open the document in the browser and send editing requests via APIs. The application authenticates the user and checks permissions. Changes are saved in the document service and stored in distributed cloud storage with version control. Other users receive real-time updates through APIs. Notifications are sent for activities like edits and sharing.

5. Benefits :

- Real-time collaboration
- Version history and recovery
- Access from anywhere, any device
- Highly available and scalable
- Secure and reliable storage

6. Security :

- Data is encrypted in transit and at rest
- IAM for user authentication and authorization
- Role-based access control for documents
- Secure APIs using HTTPS
- Permissions control for sharing and editing

7. Challenges / Trade-offs :

- Latency in real-time synchronization
- Storage cost for multiple versions
- Complex access control in large teams
- Ensuring global availability
- Data privacy and compliance

8. Conclusion :

The cloud-based document collaboration system enables users to create, edit, and share documents in real time. By using cloud storage, web APIs, and secure platform services, it ensures high availability, scalability, and security for enterprise collaboration.

Case Study 3 : Online Food Ordering Platform

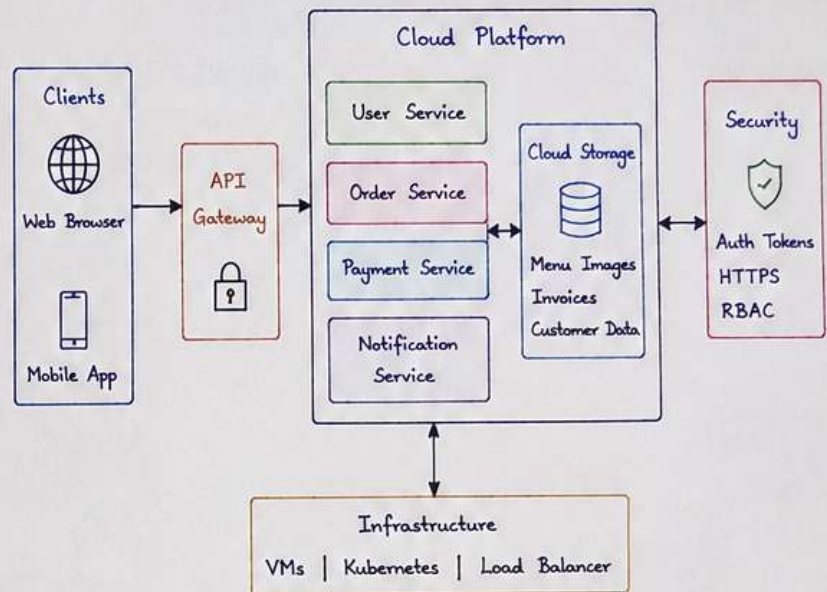
1. Introduction :

A startup builds a cloud-based food ordering platform accessible via web & mobile clients. It uses APIs, cloud storage and platform services for scalable and secure operations. The system allows customers to browse menus, place orders, make payments and track orders in real time.

2. Cloud Components Used :

- Clients : Web, Mobile Apps
- Web APIs : RESTful APIs
- Cloud Storage : Menu images, invoices, customer data
- Platform Services : Payments, Notifications, User Management
- Infrastructure : VMs, Kubernetes, Load Balancer

3. Proposed Architecture :



4. Working / Data Flow :

Clients (web or mobile) send requests to the API Gateway. The gateway authenticates the request and forwards it to the appropriate microservice (User, Order, Payment or Notification). Services process the request and store or retrieve data from Cloud Storage. Payments are processed securely through the Payment Service. Notifications are sent to users for order updates. All data is stored securely in cloud storage. The infrastructure (VMs, Kubernetes and Load Balancer) ensures scalability, reliability and high availability.

5. Benefits :

- Scalable & reliable platform
- Real-time order processing & tracking
- Secure payments & data storage
- Better user experience
- Cost effective due to cloud resources
- Easy maintenance & updates

6. Security :

- Authentication using tokens (JWT)
- HTTPS for secure communication
- Role-based access control (RBAC)
- Encrypted storage for sensitive data
- Secure APIs and input validation
- Regular backups and monitoring

7. Challenges / Trade-offs :

- Handling high traffic during peak hours
- Managing cost with auto-scaling resources
- Ensuring data consistency across services
- Complexity in microservices communication
- Dependency on cloud platform & internet

8. Conclusion :

The Online Food Ordering Platform integrates clients, APIs, cloud storage and platform services to deliver a secure, scalable and real-time solution. It provides efficient food ordering experience while ensuring high availability, reliability and data security.